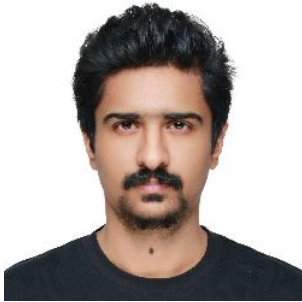




Dr. ADIL HUSSAIN

PhD in Traffic Information Engineering & Control | Electric Vehicles and Charging Station Optimization, Artificial Intelligence and Intelligent Transportation Systems

Date of birth: 1996/09/21 | **Nationality:** Pakistani | **Status:** Married

Home Address: House # PD-1415, Dhoke Babu Irfan, Stadium Road, Rawalpindi, Punjab, Pakistan.

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SUMMARY

I hold a Ph.D. in Traffic Information Engineering and Control from Chang'an University (June 2026). My research interests focus on the intersection of Artificial Intelligence (AI), deep learning, and optimization for intelligent transportation systems (ITS), electric vehicle (EV) research, and autonomous systems. I specialize in predictive modeling, scheduling, and data-driven decision-making, with recent work extending to Battery Management Systems (BMS) and Unmanned Vehicles, especially Unmanned Aerial Vehicles (UAVs) and Unmanned Surface Vehicles (USVs), in path planning, trajectory optimization, and cyber-physical security. My doctoral work was recognized with multiple awards, including the Excellent Graduate Student (2026), 2 times Star of Study Awards (2025 and 2024), Excellent International Student (2025), and the Outstanding Teaching Assistant (2025).

EDUCATION

- **Ph.D, Traffic Information Engineering and Control** 2022/09 – 2026/06

Chang'an University, Xi'an, China

Thesis Title: Research on Integrated Framework and Strategies for Electric Vehicle Charging Stations Optimization (**Supervisor:** Professor QingChang Lu)

- **Master of Science in Computer Science (MSCS)** 2019/09 – 2022/01

Bahria University, Islamabad, Pakistan

Supervisor: Dr. Kashif Naseer Qureshi

- **Bachelor of Science in Information Technology (BSIT)** 2014/09 – 2018/07

PMAS Arid Agriculture University, Rawalpindi, Pakistan

WORK EXPERIENCE

- **Teaching Assistant (Tutor)** 2024/09/01 – 2026/06/30

Academic English Writing Workshop by International Education School, Chang'an University

Delivered lectures on research-related topics, assisted other tutors during sessions, guided new students in developing early research skills, and supported students in preparing and publishing research in journals and conferences.

- **Android Developer** 2022/03/28 – 2022/11/19

Speckpro Digital, Islamabad, Pakistan

Delivered frontend development and API integration across product- and project-based Android apps, including “Salon Wizz,” the e-commerce app “Kasa Beeston,” and the “CASA Tenant Management System,” after converting a one-week trial into a full-time role.

- **Research Assistant** 2019/09/07 – 2021/09/31

Cyber Reconnaissance and Combat Lab (CRC), Islamabad, Pakistan

Conducted research in Computer Science, co-authored and developed research papers, assisted in manuscript submission and revision, and prepared publication-ready documents by updating references and ensuring formatting compliance.

- **Junior Android Developer & Intern** 2019/02/07 – 2019/09/20

O3 Interfaces (Pvt) Ltd., Islamabad, Pakistan

Converted a 3-month internship into a full-time role on the “UBL Digital” mobile banking app: implemented and tested REST APIs, performed QA and daily bug fixes, and designed and improved UI/backend features, including bill payments, card management, and digital onboarding (SSDOB).

HONOURS & AWARDS

- **2026 Excellent International Graduate Student** 2026/06

International Education School, Chang'an University, Xi'an, China

- **2025 Star of Study Award** 2025/12

International Education School, Chang'an University, Xi'an, China

- **2025 Excellent International Student** 2025/12

International Education School, Chang'an University, Xi'an, China

- **2024 Outstanding Teaching Assistant** 2025/05

International Education School, Chang'an University, Xi'an, China

- **2024 Star of Study Award** 2024/12

International Education School, Chang'an University, Xi'an, China

SKILLS & EXPERTISE

- **Research:** Scientific writing, literature review, research design, academic publishing, data analysis, dissertation supervision, teaching
- **Technical:** Machine Learning, Deep Learning, Artificial Intelligence
- **Programming:** Python, Java, C#, C++
- **Research Tools:** LaTeX, Origin, QGIS
- **Reference Management:** EndNote, Mendeley, Zotero
- **Tools & IDEs:** Android Studio, Visual Studio, PyCharm, Jupyter, Google Colab
- **Databases:** MS SQL, MySQL, SQLite
- **Productivity:** MS Office (Word, PowerPoint, Excel, Access, OneNote)
- **Versioning & PM:** Bitbucket, GitHub, JIRA, Asana
- **Design & API:** Adobe XD, Zeplin, Canva, Postman, Advanced REST Client
- **Languages:** English (Good) · Chinese (Basic) · Punjabi (Native) · Urdu (Excellent)

JOURNALS, CONFERENCES, SEMINARS & REVIEWER

- **World Transport Convention (WTC):** WTC 2024, Qingdao, China (2024/06) - Poster presentation;
- **Journals served as a Reviewer**
 - **Springer Journals:** Scientific Reports, Discover Artificial Intelligence, Journal of Big Data, International Journal of Data Science and Analytics, International Journal of Machine Learning and Cybernetics, Knowledge and Information Systems.
 - **IEEE Journals:** IEEE Transactions on Smart Grid,
 - **Elsevier Journals:** Electric Power Systems Research, Cluster Computing,
 - **Frontiers Journals:** Frontiers in Artificial Intelligence, Frontiers in Computer Science
 - **Wiley:** Expert Systems
 - **MDPI:** Sustainability, World Electric Vehicle Journal (WEVJ.)
- **Conferences served as the Technical Program Committee (TPC) & Reviewer**
 - 3rd IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2026
 - 2nd IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2025
 - 1st IEEE International Conference on Advanced Technologies in Energy and Informatics (ICATEI) 2025
 - 1st IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2024

PUBLICATIONS

Selected Publications

1. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi. [Context-Aware Temporal Fusion Transformer for Short-Term EV Charging Demand Prediction](#). [Sustainable Energy, Grids and Networks](#) (2026)
(Web of Science 2025: SCIE, Q1, IF: 5.7)
2. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi*, Khalid Javeed. [On-Demand Dynamic Charging Pricing for Electric Vehicles](#). [Egyptian Informatics Journal](#) (2026).
(Web of Science 2025: SCIE, Q2, IF: 4.2)
3. **Adil Hussain**, Qing-Chang Lu*, Sanam Shahla Rizvi, Shixin Wang, Se Jin Kwon*. [Short Term Demand Forecasting of Electric Vehicle Charging Stations using Context Aware Temporal Transformer Model](#). [Scientific Reports](#) (2025)
(Web of Science 2025: SCIE, Q1, IF: 4.9)
4. **Adil Hussain***, Vishwanath Eswarakrishnan, Ayesha Aslam, Sajib Tripura. [Charging stations demand forecasting using LSTM based hybrid transformer model](#). [Scientific Reports](#) (2025)
(Web of Science 2025: SCIE, Q1, IF: 4.9)
5. Abdullahi, Muhammad Rabiul, Qing-Chang Lu, **Adil Hussain***, Sajib Tripura, Peng-Cheng Xu, ShiXin Wang. [Location optimization of EV charging stations: A custom K-means cluster algorithm approach](#). [Energy Reports](#) (2024)
(Web of Science 2025: SCIE, Q2, IF: 6.6)

Journal Articles – Published

2026

6. Tariq, Zhu Wei, Kamran Shaukat*, Ayesha Aslam, **Adil Hussain**, Md. Mahboob Alam**. *A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning*. [Scientific Reports](#) (Accepted on 13th August 2026)
7. Sajib Tripura, Qing-Chang Lu*, Sanam Shahla Rizvi, **Adil Hussain**, Tae-Sun Chung**. [Graph Attention Network-Based Accident Risk Aware Message Dissemination for IoV Safety](#). [Expert Systems with Applications](#)
8. Mostak Ahamed, Ibrahim Kholilullah, Hairong Gu*, Md Moin Shikder, Munju Mia, Sajib Tripura, **Adil Hussain**. [Automated control integration of solar and wind energy for sustainable DC microgrid deployment in desert environments for EV charging](#). [Discover Electronics](#)

2025

9. Qing-Chang Lu*, Shixin Wang, Peng-Cheng Xu, Jing Li, Xu Meng, **Adil Hussain**. [Modeling the Dependency Relationship of Coupled Power and Transportation Network](#). [Energy](#)

10. Ayesha Aslam, Yang Xiaojun*, Kashif Naseer Qureshi**, **Adil Hussain**, Kayhan Zrar Ghafoor. [*GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction*](#). **[Journal of Engineering Research](#)**
11. Faizan Zaman, Zhigang Xu*, **Adil Hussain**, Anees Ullah, Khalid Zaman [*Enhancing Driver Emotion Recognition Through Deep Ensemble Classification*](#). **[Journal of Intelligent and Connected Vehicles](#)**
12. Sajib Tripura, Qing-Chang Lu*, **Adil Hussain**, Tanjim Mehmud. [*Optimizing traffic safety message dissemination and resource allocation using adaptive deep reinforcement learning in fog-enabled internet of vehicles network*](#). **[Egyptian Informatics Journal](#)**
13. Sajib Tripura, Qing-Chang Lu*, Dhonita Tripura, MD Ibrahim Kholilullah, Arunav Malik Avi, Mostak Ahmed, **Adil Hussain**. [*Adaptive Q-Learning-Based Event-Prioritized QoS and Incentive Optimization for Enhancing Safety in Vehicular Fog Networks*](#). **[Egyptian Informatics Journal](#)**
14. Amna Ikram*, Sunnia Ikram, El-Sayed M. El-Kenawy, **Adil Hussain**, Amal H. Alharbi*, Marwa M. Eid. [*A fuzzy-optimized hybrid ensemble model for yield prediction in maize-soybean intercropping system*](#). **[Frontiers in Plant Science](#)**

2024

15. Tariq, Vishwanath Eswarakrishnan*, **Adil Hussain**, Zhu Wei, Muhammad Uzair. [*A Novel Energy Replenishment Algorithm to Increase the Network Performance of Rechargeable Wireless Sensor Networks*](#). **[Sensors](#)**
16. **Adil Hussain***, Ayesha Aslam, Sajib Tripura, Vineet Dhanawat, Varun Shinde. [*Weather Forecasting Using Machine Learning Techniques: Rainfall and Temperature Analysis*](#). **[Journal of Advances in Information Technology](#)**
17. **Adil Hussain***, Ayesha Aslam. [*Hate Speech against Women and Immigrants: A Comparative Analysis of Machine Learning and Text Embedding Techniques*](#). **[Journal of Applied Research and Technology](#)**
18. **Adil Hussain***, Ayesha Aslam. [*Ensemble-based Approach using Inception V2, VGG-16, and Xception Convolutional Neural Networks for Surface Cracks Detection*](#). **[Journal of Applied Research and Technology](#)**
19. **Adil Hussain**, Amna Khatoon*, Ayesha Aslam, Tariq, Muhammad Asif Khosa. [*A Comparative Performance Analysis of Machine Learning Models for Intrusion Detection Classification*](#). **[Journal of Cyber Security](#)**
20. Ayesha Aslam, **Adil Hussain***. [*A Performance Analysis of Machine Learning Techniques for Credit Card Fraud Detection*](#). **[Journal of Artificial Intelligence](#)**
21. **Adil Hussain***, Ayesha Aslam. [*Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models*](#). **[Journal of Artificial Intelligence](#)**

2023

22. **Adil Hussain**, Kashif Naseer Qureshi*, Khalid Javeed, Musaed Alhussein. [*Enhanced Intelligent Intrusion Detection System for E-Commerce*](#). **Computer Systems Science & Engineering**
23. **Adil Hussain**, Asad Ullah*, Ayesha Aslam, Amna Khatoon. [*A Modified Siamese Network for Facial Assimilation*](#). **WSEAS Transactions on Signal Processing**

2022

24. Kashif Naseer Qureshi, Adi Alhudhaif, **Adil Hussain**, Saleem Iqbal, Gwanggil Jeon. [*Trust aware energy management system for smart homes*](#). **Computers & Electrical Engineering**

Journal Articles - Under Review / Revision

1. **Adil Hussain***, Ayesha Aslam, Noman Iqbal, Muhammad Yasir, Sajib Tripura. *Short-Term Charging Station Occupancy Prediction using ToU-based Demand Regime Attention*. **Applied Soft Computing** - (Under Review)
2. **Adil Hussain**, Kashif Naseer Qureshi*, Ayesha Aslam. *EV Charging Demand Prediction using Deep Learning: A Comprehensive Literature Review*. **Journal of Big Data** - (Under Review)
3. Shadab Hanifi, **Adil Hussain***, Asif Khan, Ayesha Aslam. *AI-Based Pavement Distress Assessment: A Critical Review of Sensing, Learning, and Deployment*. **Archives of Computational Methods in Engineering** (Revision)
4. Noman Iqbal, **Adil Hussain***, Ayesha Aslam, Muhammad Yasir, Lucas Sahani Ludanho. *Short-Term Traffic Congestion Forecasting using Fuzzy C-Means Clustering-based Hybrid Approach*. **Cluster Computing** (Under Review)
5. Ayesha Aslam, Xiaojun Yang*, Kashif Naseer Qureshi**, Adil Hussain. *CrossTrans-Fuse: Transformer & Cross-Attention Fusion for Lane Change Detection*. **Transportation Research: Interdisciplinary Perspectives** (Revision Submitted)
6. Muhammad Yasir, Ying Xu*, Mehboob Hussain, Adil Hussain. *An Enhanced YOLOv9-based hybrid model for Multi-Scale Acne Detection Using CSPDarknet53 with incorporation of Feature Pyramid Networks*. **Biomedical Signal Processing and Control** (Under review)

Conference Papers - Published

2026

1. **Adil Hussain***, Ayesha Aslam. [*A Reinforcement Learning based Adaptive Dynamic Pricing for EV Charging Stations*](#). **World Transport Congress (WTC)**, Tiajin, China.

2025

1. Sajib Tripura, **Adil Hussain**, Mostak Ahamed, Dhonita Tripura, Ayesha Aslam. [*A Dual Attention Mechanism-Based Enhanced Residual Network for Skin Cancer Detection*](#). **IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS)**, Bangladesh.
2. Ayesha Aslam, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, **Adil Hussain**. [*Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting*](#). **IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS)** Bangladesh.
3. **Adil Hussain**, Muhammad Yasir, Noman Iqbal, Sajib Tripura, Ayesha Aslam. [*Heart Disease Classification Using Machine Learning Techniques: An Impact Analysis of GridSearchCV-based Optimization*](#). **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
4. Ayesha Aslam, **Adil Hussain**, Noman Iqbal, Muhammad Yasir, Sajib Tripura. *Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning*. **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
5. Peng-Cheng Xu, Qing-Chang Lu, Feng Tao, Jing Li, **Adil Hussain**, Wang Rundong. *Network-Based Resilience Assessment of Metro Stations: The Relation of Infrastructures and Passengers*. **104th Transportation Research Board (TRB) Annual Meeting**, United States.

2024

6. **Adil Hussain***, Ayesha Aslam. [*A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot*](#). **World Transport Congress (WTC)** Qingdao, China.
7. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Noman Iqbal, Sajib Tripura. [*Credit Card Fraud Detection Using Machine Learning Techniques: Dealing with Imbalanced Data Using Over-Sampling and Under-Sampling Methods*](#). **IEEE Beyond Technology Summit on Informatics International Conference (BTS-I2C)** Indonesia.
8. **Adil Hussain**, Kashif Naseer Qureshi, Raja Waseem Anwar, Ayesha Aslam. [*A Novel SCD11 CNN Model Performance Evaluation with Inception V3, VGG16 and ResNet50 Using Surface Crack Dataset*](#). **IEEE 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)** Oman.
9. **Adil Hussain**, Kashif Naseer Qureshi, Faizan Zaman, Ayesha Aslam. [*Minor Surface Cracks Detection using SCD11 Convolutional Neural Network*](#). **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
10. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam, Tariq, Muhammad Rabiul Abdullahi. [*TSC18 Convolutional Neural Network for Traffic Sign Classification*](#). **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
11. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Faizan Zaman. [*Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model*](#). **IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAET)** Malaysia.

12. Amna Khatoon, Weixing Wang, Asad Ullah, Ishfaq Ahmad, **Adil Hussain**, Mardeni Roslee. [*Advancement in Pavement Condition Assessment: An AI-Based Computer Vision Approach*](#). [**IEEE Multimedia University Engineering Conference \(MECON\)**](#) Malaysia.
13. Ayesha Aslam, Sajib Tripura, Muhammad Rabiul Abdullahi, Vineet Dhanawat, **Adil Hussain**. [*Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer \(ViT\) Model*](#). [**3rd International Conference on Data Science and Emerging Technologies \(DaSet\), Springer**](#) Malaysia.
14. Faizan Zaman, Zhigang Xu, **Adil Hussain**, Ayesha Aslam, Muhammad Rabiul Abdullahi. [*Concrete and Pavement Cracks Detection Leveraging Vision Transformer \(ViT\) Model*](#). [**IEEE International Conference on Electrical and Information Technology \(IEIT\)**](#) Indonesia.

2023

15. **Adil Hussain**, Ayesha Aslam, Tariq. [*Traffic Sign Classification Using a Novel TSC18 Convolutional Neural Network*](#). [**9th International Conference of Sabaragamuwa University of Sri Lanka**](#). (Abstract) Sri Lanka.

Conference Papers - Submitted / Under Review

1. Noman Iqbal, Zahirul Islam Kajal, Sajib Tripura, **Adil Hussain**, Ayesha Aslam, Mostak Ahamed. *A Cross-Attention Enhanced BiLSTM-GAT Framework for Spatio-Temporal Traffic Flow Forecasting*. [**IEEE 3rd INTERNATIONAL CONFERENCE ON COMPUTING, APPLICATIONS AND SYSTEMS**](#) Bangladesh (2026)
2. Noman Iqbal, Zahirul Islam Kajal, Sajib Tripura, **Adil Hussain**, Ayesha Aslam, Mostak Ahamed. *Traffic Speed Prediction using LSTM and Spatial GCN-based Feature Fusion*. [**IEEE 3rd INTERNATIONAL CONFERENCE ON COMPUTING, APPLICATIONS AND SYSTEMS**](#) Bangladesh (2026)

Book Chapters

2026

1. **Adil Hussain**, Ayesha Aslam, Mahad Ahmad. [*Artificial Intelligence in Agriculture: Applications, Challenges, and Future Directions*](#) (Chapter 3). [**AI-Driven Farming Systems With Minimal Human Intervention**](#), 1st Edition: **IGI-Global**, eISBN13: 9798337392974.

2025

2. **Adil Hussain**, Ayesha Aslam, Rundong Wang, Kashif Naseer Qureshi. [*Advanced Learning Techniques for AI in Renewable Energy*](#) (Chapter 7). [**Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks**](#) 1st Edition, **Elsevier**, eBook ISBN: 9780443406195.
3. Ayesha Aslam, **Adil Hussain**, Kashif Naseer Qureshi. [*Big Data Analytics for AI in Renewable Energy Networks*](#) (Chapter 9). [**Artificial Intelligence-based Renewable Energy Systems**](#)

[Standards, Communication Systems, and Data Networks](#) 1st Edition, Elsevier, eBook ISBN: 9780443406195.

4. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam. [Novel Machine and Deep Learning and Training Techniques for AI-based Healthcare Systems](#) (Chapter 7). [Artificial Intelligence-based Smart Healthcare Systems: New Standards, Technologies and Communication Systems](#) 1st Edition, Elsevier, Paperback ISBN: 9780443264764.
5. Ayesha Aslam, Kashif Naseer Qureshi, **Adil Hussain**. [Advanced AI-Based Healthcare Systems Applications and Services](#) (Chapter 4). [Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns](#) 1st Edition. Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.
6. **Adil Hussain**, Peng-Cheng Xu, Wang Shixin, Kashif Naseer Qureshi. [Advanced AI-Based Healthcare Systems Applications and Services](#) (Chapter 6). [Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns](#) 1st Edition. Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.

2024

7. **Adil Hussain**, Kashif Naseer Qureshi. [Standards and Policies Adoption for AIoT Networks](#) (Chapter 4). [Artificial Intelligence of Things \(AIoT\): New Standards, Technologies and Communication Systems](#), 1st Edition, Taylor & Francis - CRC Press, eBook ISBN: 9781003430018.

REFERENCES

- The referee details are provided on the next page.
- Reference emails can be furnished if required.

LIST OF REFEREES

Ph.D Supervisor:

Professor Qing-Chang Lu

Chang'an Scholar Chair Professor, Vice Dean / School of Electronics and Control Engineering
Department Chair / Department of Traffic Information and Control Engineering, Chang'an University.
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***New Affiliation:** School of Management, Shanghai University, Shanghai, China.

Email: qclu@shu.edu.cn

Master's Supervisor:

Dr. Kashif Naseer Qureshi

Associate Professor, Department of Computer Science, Bahria University, Islamabad, Pakistan.

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***New Affiliation:** Department of Electronics and Computer Engineering, University of Limerick, Limerick, Ireland.

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Professor Shuguang Li

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Department of Traffic Information and Control Engineering,
Chang'an University.
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